SIM2001 Advanced Calculus : Tutorial 5

1. Change the Cartesian integral into an equivalent polar integral. Then evaluate the polar integral.

   (14.4.13) $\int_0^6 \int_0^y x dx dy$

   (14.4.15) $\int_1^{\sqrt{3}} \int_1^x dy dx$

   (14.4.17) $\int_{-1}^0 \int_{-\sqrt{1-x^2}}^0 \frac{2}{1+\sqrt{x^2+y^2}} dy dx$

   (14.4.19) $\int_0^{\ln 2} \int_0^{\sqrt{(\ln 2)^2-y^2}} e^{\sqrt{x^2+y^2}} dx dy$

   (14.4.21) $\int_0^1 \int_x^{\sqrt{2-x^2}} (x+2y) dy dx$

2. Sketch the region of integration and convert each polar integral to a Cartesian integral. Do not evaluate the integrals.

   (14.4.23) $\int_0^{\pi/2} \int_0^1 r^3 \sin\theta \cos\theta dr d\theta$

   (14.4.25) $\int_0^{\pi/4} \int_0^{2\sec\theta} r^5 \sin^2\theta dr d\theta$

3. (14.4.29)
   Find the area enclosed by one leaf of the rose $r = 12\cos 3\theta$.

4. (14.4.31)
   Find the area of the region cut from the first quadrant by the cardioid $r = 1 + \sin\theta$.

5. Evaluate the integral.

   (14.5.13) $\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2}} dz dy dx$

   (14.5.15) $\int_0^1 \int_0^{2-x} \int_0^{2-x-y} dz dy dx$

6. Find the volume of the regions. [See Textbook for the figures]

   (14.5.23) The region between the cylinder $z = y^2$ and the xy-plane that is bounded by the planes $x = 0$, $x = 1$, $y = -1$, $y = 1$

   (14.5.25) The region in the first octant bounded by the coordinate planes, the plane $y + z = 2$, and the cylinder $x = 4 - y^2$

   (14.5.29) The region common to the interiors of the cylinders $x^2 + y^2 = 1$ and $x^2 + z^2 = 1$

   (14.5.31) The region in the first octant bounded by the coordinate planes, the plane $x + y = 4$, and the cylinder $y^2 + 4z^2 = 16$

7. (14.7.35)
   Give the limit of integration for evaluating the integral $\iiint f(r,\theta,z) dz r dr d\theta$ as an iterated integral over the region that is bounded below by the plane $z = 0$, on the side by the cylinder $r = \cos\theta$, and on top by the paraboloid $z = 3r^2$.

8. Set up the interated integral for evaluating $\iiint_D f(r,\theta,z) dz r dr d\theta$.

   [See Textbook for the figures]

(14.7.37) D is the right circular cylinder whose base is the circle $r = 2\sin\theta$ in the xy-plane and whose top lies in the plane $z = 4 - y$

(14.7.39) D is the solid right cylinder whose base is the region in the xy-plane that lies inside the cardioid $r = 1 + \cos\theta$ and outside the circle $r = 1$ and whose top lies in the plane $z = 4$

(14.7.41) D is the prism whose base is the triangle in the xy-plane bounded by the x-axis and the lines $y = x$ and $x = 1$ and whose top lies in the plane $z = 2 - y$